

Fiona Victoria Stanley Jothiraj

Beaverton, OR

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EDUCATION

Oregon State University, Corvallis

2023 — 2027

Doctor of Philosophy (PhD) in Artificial Intelligence

GPA: 3.79/4.00

Area of Research: AI for Social Good, Applied AI/ML in Ecology

Honors: Outstanding Scholars Program

University of Washington Bothell

2021 — 2023

Master of Science in Computer Science and Software Engineering

GPA: 3.90/4.00

Thesis: Phoenix - A Federated Generative Diffusion Model

PSG College of Technology, India

2015 — 2019

Bachelor of Engineering in Robotics and Automation Engineering

GPA: 9.58/10.00

RESEARCH EXPERIENCE

Graduate Researcher

Oct 2023 - present

Oregon State University

Advisor: Dr. Rebecca Hutchinson

- Modeled plant-pollinator interactions across a decade of environmental change using Bayesian hierarchical dynamic occupancy models and probabilistic graphical models in R
- Investigated imperfect detection effects in species distribution models by simulating detection noise across linear and hierarchical statistical frameworks
- Conducted interdisciplinary research on species distribution modeling, demonstrating a 23% performance gain with boosted regression trees (BRTs) when habitat sampling is centered on bird locations rather than surveyor locations

AI Research Intern

June 2024/25 - Sep 2024/25

Micron Technology

Advisor: Seth Eichmeyer, Director of Product Yield Analytics & Analysis

- Awarded the **Micron Innovation Award**, marking the first step toward a patent filing
- Designed and deployed ML models to optimize semiconductor product yield through wafer defect analysis
- Built a non-parametric clustering system with rotation- and flip-invariant features, reducing manual analysis time by 90% (*AAAI'26 Workshop & Bridge Program*)
- Developed a novel anomaly scoring metric using conditional VAEs, diffusion models, and ViTs to detect multi-defect patterns in semiconductor manufacturing
- Presented project outcomes and business impact to the VP of Engineering

Graduate Researcher

June 2022 - Jan 2023

UW Bothell: Computation Behavioral Modeling (CBM) Research Lab

Advisor: Dr. Afra Mashhadi

- Conducted research on nostalgic behavior on social media using NLP and LLMs
- Built transformer models, including RoBERTa, DistilBERT, and an ensemble-based model for nostalgic tweet detection, achieving 0.96 Micro F1-Score
- Leveraged LLaMA2-70B to curate a large-scale dataset of $\approx 250K$ nostalgic English tweets

PUBLICATIONS

- [1] **Stanley Jothiraj, Fiona Victoria**, Hutchinson Rebecca, Lauren Ponisio, Julia Jones, Nicole Martinez (2025). Plant-Pollinator Interactions over more than a Decade of Environmental Change. *under review*
- [2] **Fiona Victoria Stanley Jothiraj**, Arunagiri Pandian, Seth A. Eichmeyer. (2025) DECOR: Deep Embedding Clustering with Orientation Robustness. *AAAI 2026 Bridge Program on Knowledge-guided Machine Learning*. Also selected at the *AAAI 2026 Workshop on AI to Accelerate Science and Engineering (AI2ASE)*
- [3] Shen Fang-Yu, **Stanley Jothiraj, Fiona Victoria**, Hutchinson Rebecca A, Hallman Tyler, Curtis Jenna R, and Robinson, W. Douglas. (2025). Species Distribution Model Performance Improves When Habitat

- Characterizations are Centered on Detected Individuals Instead of Observers. *Ecological Indicators*
- [4] **Stanley Jothiraj, F. V.**, & Mashhadi, A. (2024). Phoenix: A Federated Generative Diffusion Model. *Companion Proceedings of the ACM on Web Conference 2024 (WWW 2024)*
 - [5] **Stanley Jothiraj, F. V.**, Hong L., & Mashhadi, A. (2024) Nostalgia on Twitter: Detection and Analysis of a Large-Scale Dataset. *Proceedings of the Association for Information Science and Technology (ASIS&T)*
 - [6] Fang-Yu Shen, **Fiona Victoria Stanley Jothiraj**, Tyler A. Hallman, Rebecca A. Hutchinson, W. Douglas Robinson. (2024). Does species distribution model performance improve when habitat sampling is centered on bird locations instead of surveyor locations? *American Ornithological Society Annual Meeting*
 - [7] **Stanley Jothiraj, F. V.**, & Mashhadi, A. (2022). Personalized Emotion Detection using IoT and Machine Learning. *arXiv preprint*
 - [8] **Stanley Jothiraj, F. V.** (2022). Time Series Prediction for Food Sustainability. *arXiv preprint*
 - [9] Shamsaddini, Vahid, **Stanley Jothiraj, Fiona Victoria**, Chen, Mandy, & Mashhadi, Afra. (2022). Empirical Dynamic Modelling of the Multi-Source Park Visitation Data. *Data for Policy Conference*

INVITED TALKS

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| • ICCV CVAM Workshop | Creativity and Privacy in Advertising & Marketing | 2025 |
| • Decentralized AI Hackathon @ Stanford | Federated Learning for Generative Models | 2025 |
| • University of Washington | Federated Learning for Generative Diffusion Models | 2025 |
| • Flower Labs | Phoenix: The First Federated Generative Diffusion Model | 2023 |

EXPERIENCE

Graduate Teaching Assistant Jan 2026 - present
Oregon State University

- Courses: CS 464 Open Source Software & CS 493 Cloud Application Development
- Supported over 300 students in mastering open source workflows and cloud-native development practices
- Delivered feedback on assignments and led office hours to reinforce hands-on technical concepts

Data Scientist June 2023 - Oct 2023
Harvard in Tech

- Led NLP research as a thought leader in the Call for Action (CFA) team to detect unreliable news articles
- Influenced executive leadership to shape the roadmap of the CFA Data Science team
- Built transformer-based models to classify news headlines using open-source and in-house curated data
- Presented findings on fake news detection, feature engineering, and state-of-the-art NLP approaches

Machine Learning Engineer June 2019 - Mar 2020
Multicoreware Inc, India

- Designed and deployed EV Quantization logic in TensorFlow GPU for quantization-aware training and Tensorflow Lite inference of Deep Neural Networks
- Shipped open-source product to production powering Synopsys DesignWare EV Processors
- Mentored peers on quantization concepts and EV TensorFlow workflows using C++, Python, and Intel Intrinsics

Machine Learning Intern Dec 2018 - May 2019
Multicoreware Inc, India

- Developed a custom CNN model for skin cancer detection by training on gigabytes of clinical image data
- Implemented the LipSync™ API in Intel's OpenVINO via a high-level C++ inference engine for 5x speedup
- Deployed the LipSync quality control tool for OTT streaming providers using deep learning
- Demonstrated LipSync™ technology at the National Association of Broadcasters Show (NAB 2019)

AWARDS

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| • Professional Development Award - Oregon State University | 2024/2025 |
| • Micron Innovation Award - Micron Technology | 2024 |
| • Outstanding Scholars Program - Oregon State University | 2023 |
| • Virtual Scholarship - Grace Hopper Women in Computing Celebration | 2022 |
| • Academic Excellence (Ranked 1/80) - Robotics and Automation Engineering Association | 2019 |
| • Monarch of the Month - Individual contribution to TensorFlow quantization at Multicoreware | 2019 |

MEDIA

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|-----------------------------|---|------|
| • Oregon State University | AI student leverages OSU's interdisciplinary strengths | 2025 |
| • Semiconductor Engineering | Grouping Complex Wafer Defect Patterns Into Meaningful Clusters | 2025 |
| • The Hindu | App for Plus Two Students | 2015 |
| • Business Standard | School Girl Develops App for TN Engg Aspirants | 2015 |
| • India.com | School Girl Develops Application for Tamil Nadu Engg Aspirants | 2015 |

SERVICE

Reviewer - Conference & Workshop

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| • ICML Workshop on Technical AI Governance Research (TAIGR) | 2026 |
| • ICLR Workshop on Generative AI in Genomics: Barriers and Frontiers (Gen ²) | 2026 |
| • AAAI Workshop on Artificial Intelligence with Biased or Scarce Data (AIBSD) | 2026 |
| • AAAI International Conference on Web and Social Media (ICWSM) | 2025 |
| • ACL Workshop on DravidianLangTech | 2025 |
| • NeurIPS Workshop on Women in Machine Learning (WiML) | 2024-2025 |

Reviewer - Journal

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| • ACM Transactions on Knowledge Discovery from Data (TKDD) | 2025 |
| • IEEE Transactions on Mobile Computing (TMC) | 2024 |
| • IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) | 2024 |

CERTIFICATIONS

2022 Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning - Coursera
2022 Convolutional Neural Networks in TensorFlow - Coursera
2022 Natural Language Processing in TensorFlow - Coursera
2022 Introduction to Big Data - Coursera
2022 Taming Big Data with Apache Spark and Python – Udemy
2021 Amazon Web Services (AWS) - Machine Learning Specialty - Certified

SKILLS

- **Languages:** Python, R, C, C++, CUDA, MATLAB, SQL
- **A.I Tools:** TensorFlow, PyTorch, Diffusers, Transformers, OpenVINO, Keras, HuggingFace, Flower, Caffe, Scikit-learn, OpenCV, Kats, Pandas, NumPy, Jupyter, PySpark, Matplotlib, SciPy, Weights & Biases
- **R Libraries:** Dismo, Glmnet, Unmarked, DynamicSDM, Maxnet, Raster, Terra, Tidymverse, Nimble
- **Other Tools:** AWS (IoT Core, Sagemaker, Lambda, Kinesis, Glue, S3, SNS), Azure (IoT Hub, Stream Analytics, Function App), LaTeX, Git, Google Earth Engine, Docker